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Very accurate potential energy curve of the LiH molecule

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We present very accurate calculations of the ground-state potential energy curve (PEC) of the LiH molecule performed with all-electron explicitly correlated Gaussian functions with shifted centers. The PEC is generated with the variational method involving simultaneous optimization of all Gaussians with an approach employing the analytical first derivatives of the energy with respect to the Gaussian nonlinear parameters (i.e., the exponents and the coordinates of the shifts). The LiH internuclear distance is varied between 1.8 and 40 bohrs. The absolute accuracy of the generated PEC is estimated as not exceeding 0.3 cm^{-1}. The adiabatic corrections for the four LiH isotopologues, i.e., ^{7}LiH, ^{6}LiH, ^{7}LiD, and ^{6}LiD, are also calculated and added to the LiH PEC. The aforementioned PECs are then used to calculate the vibrational energies for these systems. The maximum difference between the computed and the experimental vibrational transitions is smaller than 0.9 cm^{-1}. The contribution of the adiabatic correction to the dissociation energy of ^{7}LiH molecule is 10.7 cm^{-1}. The magnitude of this correction shows its importance in calculating the LiH spectroscopic constants. As the estimated contribution of the nonadiabatic and relativistic effects to the ground state dissociation energy is around 0.3 cm^{-1}, their inclusion in the LiH PEC calculation seems to be the next most important contribution to evaluate in order to improve the accuracy achieved in this work.
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I. INTRODUCTION

Since the early work of Hylleraas on the helium atom,[1] it is common knowledge that to accurately account for the interaction between the electrons in a molecule, wave functions that explicitly depend on interelectronic distances must be employed. To overcome the computational difficulties associated with the use of the Hylleraas functions for systems with more than two to three electrons, in 1960 Boys[2] and Singer[3] introduced a simpler format of basis functions that explicitly depend on the interelectron distances, the so-called explicitly correlated (or exponentially correlated) Gaussian functions (ECGs). ECGs, due to the simplicity of calculating the Hamiltonian matrix elements with those functions, have become popular in very accurate quantum mechanical calculations of small atoms and molecules in the last 20 years. They have been successfully applied in very accurate atomic and molecular calculations performed with and without the assumption of the Born–Oppenheimer (BO) approximation for systems with various numbers of particles.[4] Those include very accurate calculations of the BO potential energy surfaces (PESs) of two-electron,[5] three-electron,[6] and four-electron[7] systems.

In recent years we have developed and implemented several methods for more efficient generation of PESs of small molecular systems employing ECGs. As computational resources become increasingly more accessible and affordable, such calculations become more feasible. However, as the size of the molecules increases, the complexity and the cost of the ECG PES calculations also increase. Let us take two-electron systems with different numbers of nuclei as an example. For a system with one nucleus, e.g., the helium atom, one can reach 1×10^{-9} hartree (Eh hereafter) accuracy of the energy with only 200 Gaussians.[8] In case of two and three nuclei, H_2 and H_3^+, 500 and 1000 Gaussians are needed, respectively, to achieve similar accuracy.[5,9] Also, for a four-electron, four-nuclear system, $(H_2)_2$, even 7000 Gaussians only yields 1×10^{-6} Eh accuracy.[7] In the ECG calculations of larger molecular systems, besides the need for larger basis sets, a problem, which one can encounter and has to deal with more often, is the occurrence of the linear dependency between the basis functions. As it is shown in this work, to effectively deal with this problem requires a rather sophisticated procedure.

In our previous works, new, better approaches to calculate PESs of small molecular systems using ECGs have been implemented and tested. Among those developments, we should especially mention the approach used in the PES calculations of the $(H_2)_2$ cluster[7] where better techniques for handling the usage of memory, for guessing new basis functions, for approximating basis sets, for new geometric configurations, and for optimizing basis sets were proposed. These techniques are now implemented in the present calculations aimed to generate a very high-accuracy potential energy curve (PEC) of the LiH molecule.

LiH is one of the smallest diatomic molecules, which has been extensively studied by spectroscopists and theoretical chemists since 1930s.[10] Highly accurate experimental data are available for this system.[11] Among the reported *ab initio* calculations of the LiH molecule, one should particularly mention the benchmark ECG calculations by Cencek and Rychlewski.[12] They evaluated BO energy of the LiH

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molecule at its experimental equilibrium internuclear distance and achieved the accuracy below 1×10^{-5} Eh with 2400 ECGs.

Though significant progress was made in Cencek and Rychlewski's calculations, there is still room for improvement. For example, a better energy minimization approach can be applied to the variational optimization of the ECGs to more effectively lower the energy while employing fewer Gaussians. Furthermore, as it is demonstrated in the present work, the whole PEC can now be calculated with a higher accuracy than the accuracy of Cencek and Rychlewski's single-point equilibrium-geometry calculation. Such a PEC enables calculations of the vibrational spectrum of LiH, which can be directly compared with the experimental data. As a matter of fact, one of the goals of the present work is to determine whether vibrational calculations carried out with the PEC we generate with ECGs are capable of delivering a subwavenumber accuracy for the whole range of the vibrational transitions of the ground electronic state of LiH.

The present calculations concern only the ground electronic state ($X^1\Sigma^+$) PEC of the LiH molecule. However, the presently implemented approach also enables excited state calculations provided that there are no nodes in the wavefunctions of those states at the nuclei. If such nodes exist, the Gaussians in the basis set need to be multiplied by a coordinate or a product of coordinates—an option which has not been implemented yet in our calculations.

The subwavenumber strive in the present calculations requires that effects beyond the BO approximation are included in the energies of the PEC. Only then one can expect to calculate the LiH vibrational energies in subwavenumber agreement with the experimental data. Thus, in the present work, for each PEC point we also calculate the adiabatic correction with the approach described in Sec. II.

II. THE METHOD

A. Wave function

The spatial part of the electronic wavefunction representing the ground state of the LiH molecule in the present calculations is a linear combination of ECGs:

$$\Psi(\mathbf{r}) = \sum_{k=1}^{M} c_k \phi_k(\mathbf{r}), \tag{1}$$

where M is the size of the basis set and c_k are the linear expansion coefficients. The ϕ_k's are the following correlated Gaussians with shifted centers, or ECGs:

$$\phi_k(\mathbf{r}) = \exp[-(\mathbf{r}-\mathbf{s}_k)'(\mathbf{A}_k \otimes \mathbf{I}_3)(\mathbf{r}-\mathbf{s}_k)]. \tag{2}$$

In Eq. (2) the symbol $\otimes$ denotes the Kronecker product, $\mathbf{I}_3$ is a 3×3 identity matrix, $\mathbf{r}$ is the $3n$ vector of the electron Cartesian coordinates, $\mathbf{s}_k$ is a $3n$ dimensional shift vector of the Gaussian centers, where n is the number of electrons, and $\mathbf{A}_k$ is a symmetric matrix. For the ECGs to be square integrable, $\mathbf{A}_k$ has to be a positive definite matrix.[13] This automatically happens if $\mathbf{A}_k$ is represented in the Cholesky-factorized form as $\mathbf{A}_k = \mathbf{L}_k\mathbf{L}_k'$, where $\mathbf{L}_k$ is a lower triangular matrix and $\mathbf{L}_k'$ is its transpose.

As the total electronic wave function must be antisymmetric with respect to the permutation of the electron labels, an appropriate symmetry projection needs to be applied to the wave function. In this work we use the spin-free formalism. Thus, the Young projection operator, $\hat{Y}$, which imposes the proper permutation symmetry, has to be applied only to the spatial part of the wave function and thus to each basis function, namely $\hat{Y}\phi_k$. $\hat{Y}$ is a linear combination of permutational operators, $\hat{P}_\gamma$, and, as the Hamiltonian is invariant with respect to all permutations of the electronic labels, in the calculation of the overlap and Hamiltonian matrix elements these operators are applied to the *ket* only. In brief, the *ket* basis functions in those matrix elements are operated on with the permutation operator $\hat{P} = \hat{Y}^\dagger\hat{Y}$ (the dagger stands for conjugate), where the $\hat{Y}$ operator can be derived using the appropriate Young tableaux representing the singlet ground state of LiH. For this state the Young operator can be chosen as: $\hat{Y}=(\hat{1}-\hat{P}_{34})(\hat{1}-\hat{P}_{12})(\hat{1}+\hat{P}_{13})(\hat{1}+\hat{P}_{24})$, where $\hat{1}$ is the identity operator and $\hat{P}_{ij}$ is the permutation of the ith and jth electrons.

B. Basis set optimization

The calculation of each PEC point is performed with the variational method and it involves optimization of three types of parameters of the wavefunction. The first two types are the nonlinear ECG parameters, i.e., the exponential $\mathbf{L}_k$ matrices and the coordinates of the Gaussian shift vectors, $\mathbf{s}_k$, as defined in Eq. (2). The third type is the linear expansion parameters, c_k, as in Eq. (1). In the variational energy minimization one can either optimize simultaneously all the above mentioned parameters or optimize them separately. Generally, as the parameters are coupled to each other, the simultaneous optimization should be more effective, but it may also lead to more frequent occurrences of linear dependencies among the ECGs. Such dependencies usually appear less often if a decoupled optimization is performed, but such an approach is more time consuming, as only one type of parameters is adjusted in this process at a time. Therefore, in our procedure one can choose to couple or decouple the optimizations of the different types of the parameters at each stage of the calculation to achieve better overall computational performance. Our calculations involve the following four stages: building the basis set for the wavefunction at the equilibrium nuclear geometry, the PEC calculations, the calculations of the adiabatic corrections at each PEC point, and the calculations of the vibrational energy levels and the corresponding transition energies.

In building the basis set, guessing of new ECGs is involved. At the beginning of the building process the initial small set of functions in our calculations is randomly chosen using Gaussian exponents taken from a standard orbital basis set. The Gaussian centers initially coincide with the nuclear centers, but, as the basis set optimization progresses, some of these centers diffuse away from the nuclei and move along the bond to the area between the two nuclei (some centers also move into the area outside the bond). After the small, initial set of functions is optimized, the calculation proceeds

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TABLE I. Comparison of the convergence of the BO energy, in Eh, for the ground state of LiH molecule at $R = 3.015$ bohrs.

Basis size	This work	Ref. 12
75	−8.068 104 2	−8.066 975
150	−8.069 654 1	−8.069 481
300	−8.070 336 2	−8.070 221
600	−8.070 494 9	−8.070 452
1200	−8.070 529 4	−8.070 512
2400	−8.070 547 3	−8.070 538
Estimated		−8.070 548[a]

[a]The estimation made in Ref. 12 using the BO energy of atoms, adiabatic corrections, and experimental equilibrium dissociation energy (see Table II for details).

to growing the basis set larger. In guessing new Gaussians, we apply the free iterative-complement interaction (FICI)[14] preoptimization procedure described in our previous works.[7] The procedure is not based on the energy minimization, but it involves optimizing the nonlinear parameters of new ECGs based on maximizing their overlap with the function generated by $(\hat{H} - E)\Psi(\mathbf{r})$ with E and $\Psi(\mathbf{r})$ calculated using the basis set already generated at that particular stage of the optimization. The advantage of the FICI procedure is that it allows for adjusting the Gaussian centers, particularly for moving them from the area around one nucleus to the area around the other, much easier than in the procedure based on the energy minimization, where such center adjustment has to overcome energy barriers. After the basis set is enlarged in this manner to the target number of functions, a global, variational, gradient-based optimization of the basis set that involves the optimization of all ECG parameters simultaneously is performed until the norm of the analytical gradient falls below a certain assumed threshold.

C. Calculations at the equilibrium bond length

Convergence of the LiH BO energy at the equilibrium bond length of $R = 3.015$ bohrs with respect to the basis size is shown in Table I. Our results are compared with the energies obtained in the benchmark calculations of Cencek and Rychlewski.[12] As one notices, our energy for every size of the basis set employed shown in the table is consistently and significantly better than the energy of Ref. 12. For example, while our energy obtained with 2400 ECGs of –8.070 547 3 Eh is very close to the estimated exact BO energy of –8.070 548 Eh (see Table II), the benchmark energy of –8.070 538 Eh obtained with the same number of functions in Ref. 12 is still off by 1×10^{-5} Eh ($\sim$2 cm^{-1}). We attribute the better convergence of our results to the use of the analytical energy gradient and to performing the global, simultaneous optimization of all Gaussian nonlinear parameters in our calculations versus a one-function-at-a-time optimization performed without the gradient in calculations of Cencek and Rychlewski.

TABLE II. Estimation of the exact LiH BO energy at the equilibrium geometry (in Eh) and its comparison with the energy obtained in the present work.

	Estimation[a]	Present
$E_{Li}(M = \infty)$	−7.478 060	
$+E_{H}(M = \infty)$	−0.500 000	
$E_{Li+H}(M = \infty)$	−7.978 060	−7.978 059 1[b]
$+\Delta E_{ad}(^7Li)$	0.000 608	0.000 608 4[b]
$+\Delta E_{ad}(H)$	0.000 272	0.000 272 3[b]
$E_{^7Li+H}(M = \text{finite})$	−7.977 180	
$-D_e$	−0.092 438	
$E_{^7LiH}(M = \text{finite})$	−8.069 618	
$-\Delta E_{ad}(^7LiH)$	−0.000 930	0.000 929 5[c]
$\mathbf{E_{LiH}(M = \infty)}$	**−8.070 548**	**−8.070 547 3**[c]

[a]Reference 12.
[b]Calculated at $R_{Li-H} = 40$ bohrs.
[c]Calculated at $R_{Li-H} = 3.015$ bohrs.

TABLE III. Estimation of the relativistic correction to the LiH equilibrium dissociation energy in Eh. Deviation of the experimental D_0 is dropped from the estimation.

E_{Li}(non-BO)	−7.477 451 93
E_{H}(non-BO)	−0.499 727 86
E_{LiH}(non-BO, $v = 0$)	−8.066 438 77[a]
D_0(non-BO)	0.089 258 98
D_0(expt.)	0.089 257 70
ΔD_0	1.3×10^{-6}

[a]Proceeding value after Ref. 4.

D. Dealing with linear dependencies between ECGs

The occurrence of linear dependencies among the ECGs becomes unavoidable during the global optimization pro-

TABLE IV. Comparison between the energies of the vibrational levels of ^{7}LiH, in cm^{-1}, calculated with the PEC including and excluding the adiabatic correction.

v	$E_{BO}(v)$	$E_{BO+ad.}(v)$	$\Delta E(v)$
0	− 19 600.57	− 19 590.10	10.48
1	− 18 240.34	− 18 230.33	10.01
2	− 16 924.98	− 16 915.42	9.56
3	− 15 653.58	− 15 644.45	9.13
4	− 14 425.30	− 14 416.59	8.71
5	− 13 239.37	− 13 231.07	8.30
6	− 12 095.19	− 12 087.28	7.91
7	− 10 992.19	− 10 984.66	7.53
8	− 9930.00	− 9922.83	7.17
9	− 8908.41	− 8901.59	6.82
10	− 7927.45	− 7920.95	6.50
11	− 6987.43	− 6981.22	6.21
12	− 6088.98	− 6083.05	5.93
13	− 5233.15	− 5227.48	5.67
14	− 4421.65	− 4416.22	5.43
15	− 3656.91	− 3651.69	5.22
16	− 2942.32	− 2937.32	5.00
17	− 2282.68	− 2277.94	4.74
18	− 1684.62	− 1680.22	4.40
19	− 1156.46	− 1152.53	3.93
20	− 708.64	− 705.34	3.30
21	− 357.53	− 355.14	2.39
22	− 118.21	− 116.83	1.38

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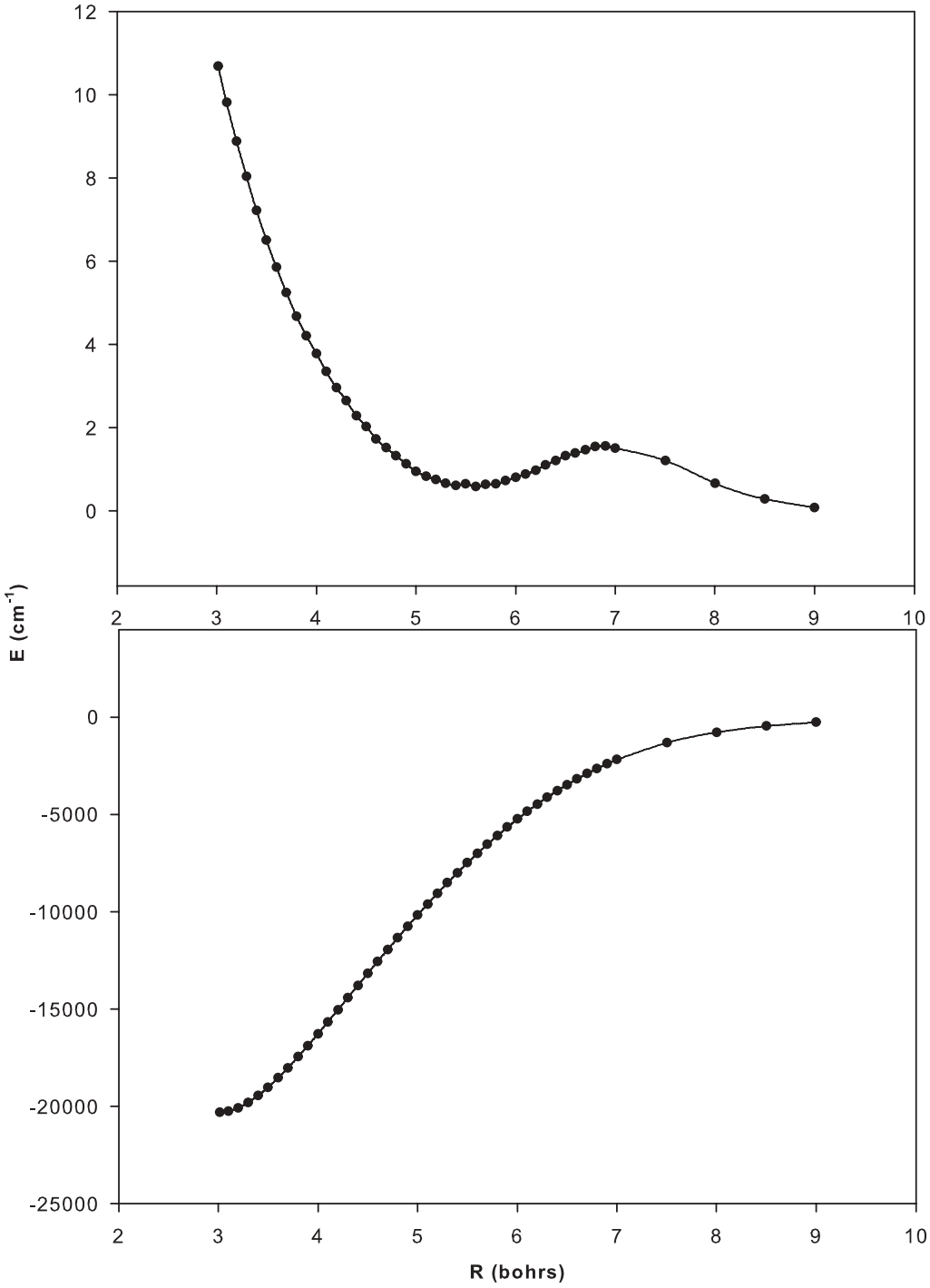


FIG. 1. The adiabatic correction (upper curve) on the total BO energy (lower curve) determined with respect to the corresponding values at $R_{\mathrm{Li-H}}$=40 bohrs as a function of the internuclear distance for the ^{7}LiH molecule.

cedure. Therefore we have developed four procedures to deal with this issue. These procedures, which we call *to distinguish*, *to replace*, *to avoid*, and *to bypass* are as follows.

1. *To distinguish:*[15] At various stages of the calculation we check the overlap of each pair of basis functions. If the absolute value of the overlap is close to one and higher than a certain assumed threshold, this pair of functions is marked as a linearly dependent and further treatment is applied to resolve the problem.

2. *To replace*:[5] It is sufficient to replace one of the functions in the linearly dependent pair to remove the linear dependency. The nonlinear parameters of the replacement function are generated by maximizing the overlap between the function and the linear combination of the functions of the linearly dependent pair taken with the linear coefficients with which the pair enters the wave function. The linearly dependent pair is replaced by a pair comprising the newly generated function and one of the old functions of the pair. After the replacement, the optimization of the basis set is restarted.

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Unfortunately, the replacement uses some extra computer time and usually results in the total energy increasing somewhat. The additional time needed for the energy to return to the value before the replacement considerably slows down the optimization process.

3. *To avoid*:[7,13] For larger basis sets, instead of replacing linearly dependent pairs, a method that prevents the formation of the linear dependencies all together in the optimization process is preferred. This is accomplished by adding penalty terms to the variational energy functional. When the value of the overlap between a pair of basis functions reaches a certain threshold, the penalty term for that pair, which was zero below the threshold, becomes positive and its value increases if the pair becomes more linearly dependent. In the minimization of the energy functional, which includes the penalty term for each pair of basis functions, the basis functions automatically stay linearly independent. We apply this method in the calculations when the basis set includes more than 300 ECGs.

4. *To bypass*:[7] In the case when the linear dependency occurs too frequently, and none of the above procedures is able to correct the problem (this happens when the basis set has around 500 ECGs) a decoupled approach is applied in the optimization. In this approach, the basis set is partitioned into subsets and each subset is optimized separately. This lowers the probability of the linearly dependency to occur. In the extreme case, each function is optimized separately. This actually allows for increasing the efficiency of the calculation by having each processor carry out optimization of a different function. The higher efficiency results from the reduction of the frequency of the interprocessor communication events (see Ref. 7 for details).

The above procedures have been involved to grow the basis set to the size of about 900 ECGs. After that there was no need to remove any linear dependencies because none appeared. At that point the basis set was nearly saturated and the optimization procedure stopped making basis functions linearly dependent to describe certain missing features of the wavefunction, because all the needed features had been, for the most part and consistently with the machine precision of the used computers, already described. This explanation is based on an observation that certain features of the wavefunction can be either represented by pairs of almost linearly dependent functions or alternatively by functions which are not linearly dependent but whose centers are shifted to the right places. It is unpredictable which representation is used in the basis set optimization. If, however, the representation with a linearly dependent pair of functions is selected, it can be converted to the other representation. This is what the *replace* procedure does. The optimization of the basis set has been carried out until it reached the target number of 2400 functions.

E. Calculation of PEC

The PEC calculations have been performed with 2400 ECGs. Rather than generating a new basis set from scratch for each PEC point, the basis set is approximated based on the basis set taken for a neighboring PEC point and reoptimized. The computer time needed for this reoptimization is dependent on how well the basis set is approximated. In generating the approximate basis, we use a procedure based on the Gaussian product theorem.[16] It allows to readjust the Gaussian centers of ECGs from one PEC point to a neighboring point. The approach works best if the points are close to each other. Hence in the present PEC calculations we use the step of 0.1 bohr for the internuclear distances shorter than 7 bohrs. Beyond this point a larger step is used. In generating an approximate basis only the Gaussian centers are adjusted, but not the $\mathbf{L}_k$ exponential parameters. The $\mathbf{L}_k$ parameters are adjusted by the reoptimization of the basis performed soon after the shifting of the ECG centers has occurred.

It is worth noting that there are as many as 14 nonlinear parameters in each Gaussian and 33 600 of those parameters in the basis set of 2400 basis functions. Only the use of the analytical gradient allows for an efficient optimization of such a large number of parameters. Also, decoupling the optimization of the Gaussian centers from the optimization of the $\mathbf{L}_k$ exponential parameters is a helpful approach. Such decoupling is based on the fact that the formula for the first derivatives of the total energy with respect to the Gaussian centers are much simpler than the formulas for the derivatives with respect to the $\mathbf{L}_k$ exponential parameters.[17] In the decoupled approach we usually perform the optimization of the Gaussian centers first.

When the internuclear distance is longer than 7 bohrs, the energy change with respect to variation of the geometric configuration becomes smaller and a larger step of 0.5 bohr is used. The LiH molecule has ionic character ($Li^+ + H^-$) at the equilibrium internuclear distance, but covalent ($Li + H$) when it dissociates. Thus, at PEC points near the equilibrium some Gaussian centers are shifted to the area between the nuclei, but as the molecule stretches these centers gradually shift back to the nuclei. This gradual change of the positions is effectively done by the above-described procedure, which works the best when the step size is small. However, due to a reason related to the center-shifting algorithm, when a large step is used, some Gaussian centers (particularly those positioned close to the coordinate center located in the middle of the bond in the present calculations) are not shifted as much as they should. Though those Gaussian centers are eventually adjusted by the basis set reoptimization, the too small initial shifting leads to some computational inefficiency. Thus, for those centers, to have them better follow the shifts of the nuclei, a different shifting procedure is applied. Namely their shifts are set equal to the shift of the nucleus they are closer to. Such shifts provide a better guess for the new positions of the Gaussian centers as the nuclei are moved apart in the PEC calculation.

F. Adiabatic correction

To partially correct for the shortcomings of the Born–Oppenheimer approximation in this work, the adiabatic correction is added to the BO energy at each PEC point. The

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TABLE V. Comparison of the calculated and experimental transition energies of ^{7}LiH in cm^{-1}. The calculated transitions include those obtained with and without the adiabatic correction. ΔE_1 and ΔE_2 are the difference between experimental values and the two types of the calculated transitions.

$v' \rightarrow v''$	Present without ad. corr.	Present with ad. corr.	Expt.[a]	ΔE_1	ΔE_2
1→0	1360.24	1359.77	1359.71	0.53	0.06
2→1	1315.36	1314.90	1314.89	0.47	0.01
3→2	1271.40	1270.97	1270.89	0.51	0.08
4→3	1228.28	1227.86	1227.77	0.51	0.09
5→4	1185.92	1185.51	1185.44	0.48	0.07
6→5	1144.18	1143.80	1143.77	0.41	0.03
7→6	1103.00	1102.62	1102.60	0.40	0.02
8→7	1062.19	1061.83	1061.78	0.41	0.05
9→8	1021.58	1021.24	1021.17	0.41	0.07
10→9	980.96	980.64	980.52	0.44	0.12
11→10	940.02	939.73	939.62	0.40	0.11
12→11	898.45	898.17	898.09	0.36	0.08
13→12	855.83	855.57	855.48	0.35	0.09
14→13	811.50	811.26	811.19	0.31	0.07
15→14	764.74	764.53	764.44	0.30	0.09
16→15	714.59	714.37	714.22	0.37	0.15
17→16	659.64	659.38	659.27	0.37	0.11
18→17	598.06	597.72	597.89	0.17	−0.17
19→18	528.16	527.69	527.91	0.25	−0.22
20→19	447.82	447.19	446.61	1.21	0.58
21→20	351.11	350.20	350.99	0.12	−0.79
22→21	239.32	238.31	237.71	1.61	0.60

[a]Reference 11.

TABLE VI. Comparison of the present pure vibrational transitions, in cm^{-1}, of ^{7}LiH in the ground $X^1\Sigma^+$ state with the results obtained by others. The values in parentheses are the differences between the calculated and the experimental values.

$v' \rightarrow v''$	Present[a]		Li and Paldus[b]		Lundsgaard and Rudolph[c]		Gadéa and Leininger[d]		Bande *et al.*[e]		Expt.[f]
1→0	1359.77	(0.06)	1361.5	(1.8)	1359.66	(−0.05)	1359.98	(0.27)	1356.87	(−2.84)	1359.71
2→1	1314.90	(0.01)	1315.4	(0.5)	1314.68	(−0.21)	1315.22	(0.33)	1312.79	(−2.10)	1314.89
3→2	1270.97	(0.08)	1271.1	(0.2)	1270.55	(−0.34)	1271.87	(0.98)			1270.89
4→3	1227.86	(0.09)	1228.3	(0.5)	1227.31	(−0.46)	1228.68	(0.91)			1227.77
5→4	1185.51	(0.07)	1186.6	(1.2)	1184.87	(−0.57)	1185.69	(0.25)			1185.44
6→5	1143.80	(0.03)	1145.7	(1.9)	1143.06	(−0.71)	1143.27	(−0.50)			1143.77
7→6	1102.62	(0.02)	1105.1	(2.5)	1101.72	(−0.88)	1101.59	(−1.01)			1102.60
8→7	1061.83	(0.05)	1064.8	(3.0)	1060.73	(−1.05)	1060.62	(−1.16)			1061.78
9→8	1021.24	(0.07)	1024.6	(3.4)	1019.88	(−1.29)	1020.24	(−0.93)			1021.17
10→9	980.64	(0.12)	984.2	(3.7)	978.85	(−1.67)	979.98	(−0.54)			980.52
11→10	939.73	(0.11)	943.4	(3.8)	937.40	(−2.22)	939.13	(−0.49)			939.62
12→11	898.17	(0.08)	901.9	(3.8)	895.21	(−2.88)	897.48	(−0.61)			898.09
13→12	855.57	(0.09)	859.2	(3.7)	851.75	(−3.73)	854.93	(−0.55)			855.48
14→13	811.26	(0.07)	814.5	(3.3)	806.39	(−4.80)	811.01	(−0.18)			811.19
15→14	764.53	(0.09)	767.1	(2.7)	758.32	(−6.12)	764.63	(0.19)			764.44
16→15	714.37	(0.15)	715.9	(1.7)	706.47	(−7.75)	713.71	(−0.51)			714.22
17→16	659.38	(0.11)	659.3	(0.0)	649.46	(−9.81)	656.18	(−3.09)			659.27
18→17	597.72	(−0.17)	595.5	(−2.4)	585.50	(−12.39)	589.90	(−7.99)			597.89
19→18	527.69	(−0.22)	521.9	(−6.0)	512.30	(−15.61)	513.45	(−14.46)			527.91
20→19	447.19	(0.58)	435.2	(−11.4)	427.12	(−19.49)	429.30	(−17.31)			446.61
21→20	350.20	(−0.79)	330.8	(−20.2)	326.95	(−24.04)					350.99
22→21	238.31	(0.60)	201.6	(−36.1)	209.30	(−28.41)					237.71
D_e(ev)	2.515 4		∼2.500 6		2.492		2.512		2.521		2.515 4

[a]This column is retaken from Table V.
[b]Multireference CCSD-[4R] stands for the coupled-cluster method with singles and doubles using a four-reference model space/cc-pVQZ (Ref. 27).
[c]Full CI/all-electron (Ref. 26).
[d]All-electron CCSD(T) (Ref. 28).
[e]FC LSE (Ref. 29).
[f]References 10 and 11.

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correction is calculated using the following Born–Handy formula:[18]

$$E_{ad} = \langle\Psi| - \frac{1}{2}\sum_{\alpha=1}^{K}\frac{1}{M_\alpha}\sum_{i_\alpha=1}^{3}\frac{\partial^2}{\partial Q_{i_\alpha}^2}|\Psi\rangle, \quad (3)$$

where M_α is the mass of nucleus α (12 786.393m_e for the ^{7}Li nucleus, 10 961.898m_e for the ^{6}Li nucleus, 1836.152 672 21m_e for the H nucleus, and 3670.482 965 4m_e for the D nucleus,[19] where $m_e = 1$ is the electron mass), Q_{i_α} is the ith coordinate of the nucleus α, and K is the number of nuclei in the system. In Eq. (3), instead of directly differentiating the electronic wave function with respect to the nuclear coordinate Q_{i_α}, we follow the approach proposed by Cencek and Kutzelnigg to approximate the derivative numerically[16] as

$$\frac{\partial\Psi}{\partial Q_{i_\alpha}} \simeq \frac{\Psi\left(Q_{i_\alpha} + \frac{1}{2}\Delta Q_{i_\alpha}\right) - \Psi\left(Q_{i_\alpha} - \frac{1}{2}\Delta Q_{i_\alpha}\right)}{\Delta Q_{i_\alpha}}. \quad (4)$$

The Gaussian centers for the wave functions at $Q_{i_\alpha} \pm \frac{1}{2}\Delta Q_{i_\alpha}$, $\Psi(Q_{i_\alpha} \pm \frac{1}{2}\Delta Q_{i_\alpha})$, are determined from their values in the $\Psi(Q_{i_\alpha})$ wave function by the same procedure as used in the PEC calculations at the internuclear distances smaller than 7 bohrs. The $\mathbf{L}_k$ parameters are not changed and the linear expansion coefficients are recalculated using the variational method. For $\Delta Q_{i_\alpha} = 1 \times 10^{-4}$ such an approach gives sufficient accuracy in the calculation of the adiabatic correction.

III. RESULTS AND DISCUSSION

As mentioned, the goal of the present work is to employ ECGs to generate the BO PEC including the adiabatic corrections for LiH that, when used in the calculations of the vibrational transition, gives the values of these transitions within a wavenumber from the experimental values. Below we describe how this goal has been accomplished.

Let us first evaluate the accuracy of the PEC we determine in the present calculations. Cencek and Rychlewski estimated the BO energy of the LiH molecule at its equilibrium internuclear distance based on accurate theoretical results for the lithium and hydrogen atoms and the experimental data.[12] We summarize their procedure and compare their results with our energies in Table II. In their calculation they first add the infinite-mass energies of the Li[20] and H atoms and their respective adiabatic corrections to get an estimate of the finite-mass LiH energy at the infinite internuclear distance. In the second step, they subtract the experimental equilibrium dissociation energy of the LiH molecule from the above value to get the finite-mass LiH energy at equilibrium. Finally, they subtract the LiH adiabatic correction from this energy to get the BO energy. This estimation provides two check points for the BO PEC, the BO energies of the LiH molecule at the equilibrium and infinite internuclear distances. Besides the BO energy, it also verifies the accuracy of the adiabatic correction at the infinite internuclear distance. We estimate that the absolute accuracy in reproducing these three values by the present calculations is around 1×10^{-6} Eh.

As the experimental dissociation energy comprises all effects including the nonadiabatic effect, the relativistic effects, etc., Cencek and Rychlewski suggested that the relativistic correction needs to be included in the calculations. Two theoretical calculations[21,22] suggested that the contribution of the relativistic correction to the equilibrium dissociation energy is around 1×10^{-5} Ehs. This is a rather significant contribution in comparison to the accuracy we aim to achieve in this work. However, as we argue below, the contribution is likely to be smaller.

Though relativistic contribution to the equilibrium dissociation energy cannot be directly determined from the results shown in Tables I and II, we believe this contribution is smaller than 1×10^{-5} Ehs. In Table III we show a procedure to estimate this contribution. The approach involves comparing

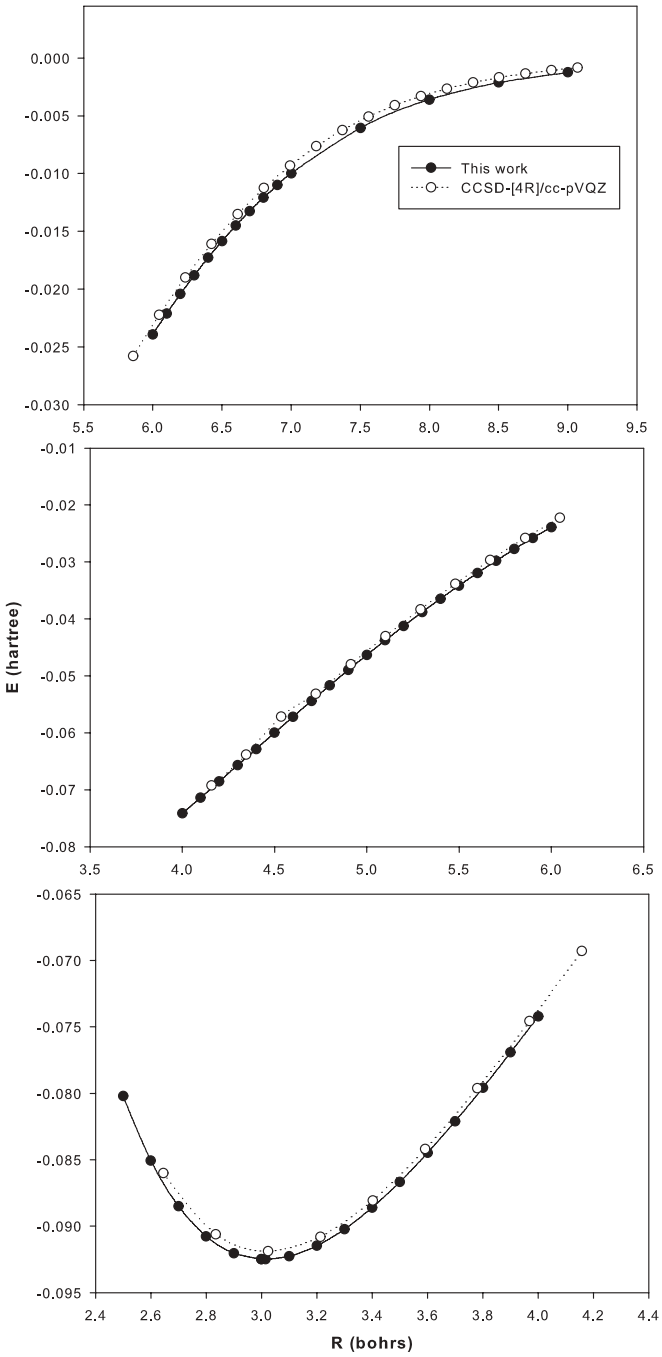


FIG. 2. Ground state BO PEC for the LiH molecule compared with the CCSD-[4R]/cc-pVQZ PEC of Li *et al.* (Ref. 27).

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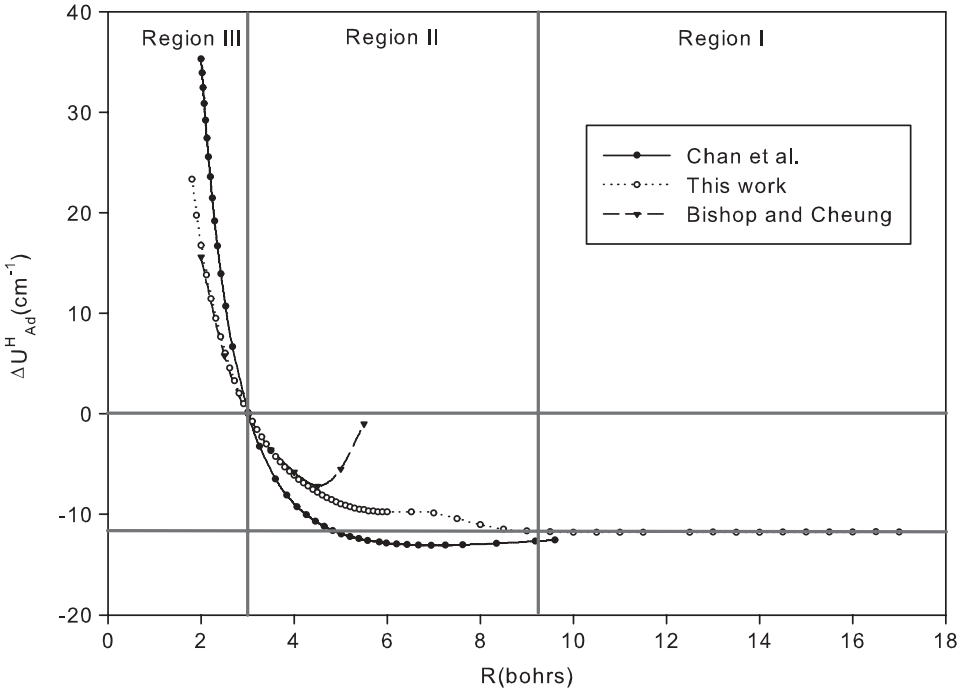


FIG. 3. Comparison of the adiabatic correction of ^{7}LiH as a function of the internuclear distance obtained in the present work with those obtained in the calculations by Bishop and Cheung (Ref. 31) and in the experimental work of Chan *et al.* (Ref. 11).

the very accurate nonrelativistic non-BO ground-state dissociation energy (D_0) of ^{7}LiH[4] with the experimental D_0[10] of this system. The difference between these two values accounts for the contribution of the relativistic correction to D_0. At $R_{\mathrm{Li-H}} = \infty$ the nonrelativistic non-BO energy of the LiH molecule is given by summing the energy of ^{7}Li[23] and H atoms. The nonrelativistic D_0 can be obtained by subtracting the ground-state nonrelativistic non-BO energy of the LiH molecule from this value. This gives the contribution of the relativistic correction to the D_0 of around 1×10^{-6} Eh. As this contribution can be expected to gradually rise in the PEC points from the equilibrium distance to the infinite distance, it is safe to assume that it does not have a significant effect on the calculation of the vibrational energy levels and the transition energies between them.

We have determined the energies of 89 PEC points in the present calculations.[24] The energy of each point includes the adiabatic correction. The correction was calculated for different isotopologues of LiH. Next we have used Le Roy's LEVEL 8.0 program[25] to calculate the vibrational energy levels and the corresponding transition energies for the different isotopologues.

The present results allow for estimation of how much the inclusion of the adiabatic correction improves the energies of the vibrational transitions. In Table IV we investigate this effect for the ^{7}LiH molecule. As the adiabatic correction shallows the PEC well, the vibrational levels calculated without the correction have lower energies (with respect to the energy at dissociation) than the vibrational levels calculated with the correction. As the adiabatic correction contributes around 10.7 cm^{-1} to the depth of the PEC well and less for stretched geometries, its impact is higher on the energies of lower levels than of higher levels. While the lowest level is raised by about 10.48 cm^{-1} by including the adiabatic correction, the highest level is raised by only 1.38 cm^{-1}.

It is interesting to see how the adiabatic correction varies with the internuclear distance. We show this behavior in Fig. 1. As one can see, the correction increases sharply at smaller distances, at intermediate distances it passes through a minimum and a maximum, and then goes to zero at larger distances.

Transition energies between adjacent vibrational levels can be more directly compared with the experimental results than the absolute energies of the levels. Therefore, in Table V we show the improvement in the values of the vibrational transitions due to including the adiabatic correction. The transitions calculated with the PEC without the correction overestimate the experimental values for all transitions. The lower transitions are more overestimated than the top transitions. This reflects the known property of the BO PECs having too steep slopes in the asymptotic regions.[26]

As one can see in Table V, the inclusion of the adiabatic correction makes a significant improvement in the transition energies. The energies become much closer to the experimental energies particularly for v smaller than 16. Above $v = 16$ the differences increase by never exceed a wavenumber.

In Table VI we compare our transition energies with the results of four previous calculations.[26–29] The table also includes the values of D_e (well depth), which is a good indicator of the accuracy of the calculations.

There is a common feature in the vibrational transition energies obtained in first two calculations. For v smaller than 10 the results agree with the experimental data within a few wavenumbers, but after that much larger deviations occur. Clearly, our transition energies are significantly closer to the experimental values than the results of the other calculations.

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TABLE VII. Comparison of the calculated and the experimental D_e and D_0 for four isotopologues of LiH (expressed in cm^{-1}). ΔE_1 and ΔE_2 are the differences between the calculated and the experimental value of D_e and D_0, respectively. Experimental deviations are not shown in ΔE_1 and ΔE_2.

Isotopomer	Present D_e	Present D_0	Expt.[a] D_e	Expt.[a] D_0	ΔE_1	ΔE_2
^{7}LiH	20 288.1	19 590.1	20 287.7±0.3	19 589.8±0.3	0.4	0.3
^{7}LiD	20 294.0	19 769.2	20 293.5±0.3	19 768.8±0.3	0.5	0.4
^{6}LiH	20 288.3	19 583.1	20 287.7±0.4	19 582.6±0.4	0.6	0.5
^{6}LiD	20 294.2	19 759.8	20 293.5±0.4	19 759.2±0.4	0.7	0.6

[a]Reference 10.

For D_e, while our results is essentially exact, the results of the other calculations differ from the experiment by several wavenumbers.

To explore the source of these deviations, we focus on the PEC of Li *et al.*,[27] as their D_e is the closest to ours and also to the experimental value. A comparison of our BO PEC with that of Li *et al.* is shown in Fig. 2. While the two curves are nearly identical, a more detail analysis shows a slight difference in the region between $R_{Li-H} = 6$ and 9 bohrs. This internuclear distance corresponds to the interaction energies in the range between 8000 and 2500 cm^{-1}. Vibrational energy levels larger than $v = 10$ are in this range. This may explain why larger deviations occur for the vibrational transitions corresponding to v larger than 10. It also indicates that maintaining high accuracy of the PEC in this region is important for getting good agreement with the experiment for the transitions at the top part of the spectrum. Clearly the ECG calculations can deliver such accuracy.

TABLE VIII. Comparison of the present pure vibrational transitions, in cm^{-1}, of ^{7}LiD, ^{6}LiH, and ^{6}LiD isotopologues in the $X^1\Sigma^+$ ground state. The values of ΔEs are the differences (Present-Expt.) between the calculated and the experimental values (Ref. 11).

$v' \rightarrow v''$	ΔE(^{7}LiD)	ΔE(^{6}LiH)	ΔE(^{6}LiD)
1→0	0.07	0.06	0.06
2→1	−0.01	0.01	−0.02
3→2	0.02	0.08	0.02
4→3	0.05	0.09	0.05
5→4	0.07	0.06	0.07
6→5	0.08	0.03	0.06
7→6	0.02	0.02	0.03
8→7	0.00	0.05	−0.01
9→8	−0.02	0.08	−0.02
10→9	0.00	0.12	
11→10	0.00	0.10	
12→11	0.02	0.10	
13→12	0.06	0.08	
14→13	0.08	0.07	
15→14	0.09	0.10	
16→15	0.12	0.17	
17→16	0.15	0.09	
18→17	0.17	−0.21	
19→18	0.19	−0.07	
20→19	0.20	0.54	
21→20	0.24	−0.86	
22→21	0.24		
23→22	0.15		
24→23	−0.12		
25→24	−0.31		
26→25	0.38		
26→26	0.16		

An additional point needs to be made concerning the comparison shown in Table VI. The recent calculations of the BO PEC of LiH performed with the free-complement (FC) local Schrödinger equation (LSE) method by Nakatsuji *et al.*[30] and Bande *et al.*[29] an absolute accuracy better than 1×10^{-4} Eh throughout the whole PEC was achieved. D_e was also determined with the same accuracy in those calculations. However, as the authors revealed, there were some fluctuations in their PEC due to the use of the Metropolis–Monte Carlo sampling, which, as one may assume, prevented them from calculating all but the lowest two vibrational levels. Those two levels are shown in Table VI.

There was a discrepancy concerning the adiabatic correction curve for the ^{7}LiH molecule between the theoretical calculations of Bishop and Cheung[31] and the experimental work of Chan *et al.*,[11] which in view of the present results we can try to explain. The adiabatic correction curve obtained in the present calculations and those of Bishop and Cheung and Chan *et al.* are compared in Fig. 3. Also here we subdivide the PECs plot into three regions. The most pronounced difference among the three PECs can be seen in Region II—the region of the avoided crossing between ionic and covalent adiabatic states. In this region, the theoretical calculations by Bishop and Cheung suggest that the curve should have an upward trend. However, according to the experimental results of Chan *et al.*, it should become flat.

A simple physical argument can be given in support of a hump in the adiabatic correction in the avoided crossing region. Because the ground-state wave function at the avoided crossing changes character (as mentioned, as the bond between Li and H stretches, it transitions from ionic to covalent) its derivative with respect to the bond-stretching coordinate should be large. Similarly, the adiabatic correction to the BO energy should also be larger in this region (see Ref. 32 for more details on this point). Thus, after initially declining with the increasing internuclear distance, the adiabatic correction should increase in the avoided crossing region before declining again at larger distances. This behavior gives rise to the hump.

Comparison of the PECs in Region III shows that our curve agrees better with Bishop and Cheung's curve than with the experimental one. Bishop and Cheung suggested that the experimental curve in this region may contain some larger nonadiabatic contributions that cannot be separated out in the experimental data.[11]

The significance of the nonadiabatic and the relativistic effects can also be observed in Table VII. Let us take ^{7}LiH as an example. The difference between the experimental and the calculated ground state dissociation energy, D_0, is 0.3 cm^{-1}. In estimating the nonadiabatic effect, we compare the present D_0 value with the estimated nonrelativistic non-BO ^{7}LiH ground state dissociation energy of 19 590.08 cm^{-1} (see Table III). The 0.02 cm^{-1} difference between the present adiabatic value and the value obtained in the nonadiabatic calculations accounts for the nonadiabatic effect.

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The remaining 0.28 cm^{-1} should account for the relativistic effects.

Finally, in Tables VI and VIII we compare the vibrational transitions calculated in this work for ^{7}LiD, ^{6}LiH, and ^{6}LiD with the experimental transitions. One can see that, similarly to ^{7}LiH, the calculated and the experimental results agree very well. The agreement is again somewhat better in the lower part of the spectrum than for the upper part, however, generally the deviation never exceeds 0.9 cm^{-1}.

IV. SUMMARY

In this work we show significantly improved calculations of the pure vibrational spectra of ^{7}LiH, ^{7}LiD, ^{6}LiH, and ^{6}LiD in their ground $X^1\Sigma^+$ electronic state performed with the use of the new LiH PEC obtained in the ECG calculations. When the PEC energies do not include the adiabatic corrections, the vibrational transitions agree with the experimental values within 1.7 cm^{-1}. Including the corrections brings the discrepancy down to below a wavenumber (to about 0.8 cm^{-1}). However, for the lowest 18 transitions in the ^{7}LiH spectrum the discrepancy is much lower than that and never exceeds 0.2 cm^{-1}. Similar accuracy is also obtained for the vibrational transitions of ^{7}LiD, ^{6}LiH, and ^{6}LiD. It is clear that one needs to achieve very high accuracy in the BO calculations of the PEC points, as well as in the calculations of the adiabatic correction, to bring the calculated frequencies this close to the experimental values.

In the BO calculations performed in this work, we have employed explicitly correlated Gaussian functions with shifted centers. Variational optimization of the nonlinear parameters of these functions with a procedure that employs the analytical energy gradient determined with respect to these parameters allows for achieving the needed accuracy of the calculations. A systematic procedure for generating extended ECG basis sets has been employed in the calculations. Though only the ground electronic state of the LiH molecule is calculated, the procedure can also be used to calculate excited states.

A question that arises: how can one increase the accuracy of the present calculations? Should the BO calculations be improved by increasing the size of the basis set, or should more post-BO effects be included? One can definitely achieve a higher accuracy in the calculation of the adiabatic correction with higher numerical precision. However, as we mentioned before, the contribution of the relativistic effects to the ^{7}LiH dissociation energy is about 1.3×10^{-6} Eh, and these effects may need to be calculated. Also, improving the PEC accuracy beyond 1×10^{-6} will likely require inclusion of the nonadiabatic corrections. These corrections are particularly important for determining the higher vibrational levels located in the energy range near the dissociation where more nonadiabatic state mixing can be expected to occur. Work in this direction will be pursued.

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